

An introduction to Python language

Constructor
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Control Flow

Subjects in this lecture

- ❑ Relational and logic Operators
- ❑ False and True in python
- ❑ If...elif ...else statement
- ❑ range() statement
- ❑ for ... in ...
- ❑ While ...
- ❑ continue and break

Relational and Logical Operators

■ Relational operators

==	equal
!=, <>	not equal
>	greater than
>=	greater than or equal
<=	less than or equal
<	less than

■ Logical operators

and	and
or	or
not	not

Control Flow

False and True in python

Things that are False

- The Boolean value False
- The numbers 0 (integer), 0.0 (float) and 0j (complex).
- The empty string "".
- The empty list [], empty dictionary {} and empty set set().

Things that are True

- The Boolean value True
- All non-zero numbers.
- Any string containing at least one character.
- A non-empty data structure.

If statement

General form

```
If <conditions>:  
    statements...  
elif <conditions>  
    statements...  
....  
else:  
    statements...
```

Example: If statement

Write program to read value from user and print word “big” if the value is greater than 10:

Example: If statement

Write program to read value from user and print “big” if the value is greater than 10 and “small” if the value is less than or equal to 10:

Example: If statement

Write program to read value from user and print “big” if the value is greater than 10 and “small” if the value is less than or equal to 10:

Example: If statement with list

Write program to read the name from a user and print “you are exist” if the name is exist in a list of names otherwise will print “you are not exist.”

Example: If statement

Write program to read a value between 0 and 100 from user and print “Excellent” if the value is between or equal to [100,90], print “very Good” if the value is between or equal to [89,80], print “Good” if the value is between or equal to [79,70], print “Medium” if the value is between or equal to [69,60], print “Pass” if the value is between or equal to [59,50], print “Fail” if the value is less than 50:

If statement exercises

- ☐ Write an if statement that asks for the user's name via input() function. If the name is "Bond" make it print "Welcome on board 007." Otherwise make it print "Good morning NAME". (Replace Name with user's name)
- ☐ Write an if statement that asks for the user's name via input() function and print True if a number is even and otherwise print False.
- ☐ Write a Python program to find those numbers which are divisible by 7 and multiple of 5, between 1500 and 2700 (both included).
- ☐ Write a Python program to find those numbers which are divisible by 7 or multiple of 5, between 1500 and 2700 (both included).
- ☐ Write a Python program that accepts a word from the user and reverse it.

range() function

- ☐ The general form:
`range(start,end,step)`
- ☐ Examples:
`range(5)`
`range(1,5)`
`range(1,10,2)`

For statement

for statement is very flexible (here first form of for):
for variables in iterable:

For statement

for statement is very flexible (here little examples):

```
for k in [1,2,3,4,9] ← list
```

```
    print (k)
```

```
for k in range(5):
```

```
    print (k)
```

```
for k in range(3,9,2)
```

```
    print (k)
```

```
for k,m in List_of_tuples
```

```
    print (k,m)
```

```
for k,(m,n) in List_of_tuples_of_tuple
```

```
    print (k,m,n)
```

Example: For statement

If you have list of values, write program to count the number that divisible by 3.

Example: For statement

If you have list of values, write program to select the minimum number and the maximum number.

Example: For statement

Write program that doing summation of list of numbers and print the result.

For statement General form

for variables in iterable:

statements

else:

statements

while statement

```
while <expression>:  
    statement(s)
```

```
count = 0  
while (count < 9):  
    print ('The count is:', count )  
    count = count + 1  
print ("Good bye!")
```

Example: while statement

If you have list of values, write program to count the number that divisible by 3.

Example: while statement

If you have list of values, write program to select the minimum number and the maximum number.

Example: while statement

Write program that doing summation of list of numbers print the result.

Break, continue

- Use “break” to stop the loop
- Use “continue” to stop processing the current item
- You can use one or more loop inside any another while, for or do..while loop.

```
>>> for value in [3, 1, 4, 1, 5, 9, 2]:
...     print ("Checking", value)
...     if value > 8:
...         print ("Exiting for loop")
...         break
...     elif value < 3:
...         print ("Ignoring")
...         continue
...     print ("The square is", value**2)
... 
```

Checking 3
The square is 9
Checking 1
Ignoring
Checking 4
The square is 16
Checking 1
Ignoring
Checking 5
The square is 25
Checking 9
Exiting for loop
>>>